Tonmya_Prescriptions

2026-08-09

Some simple models based on Tonmya sales

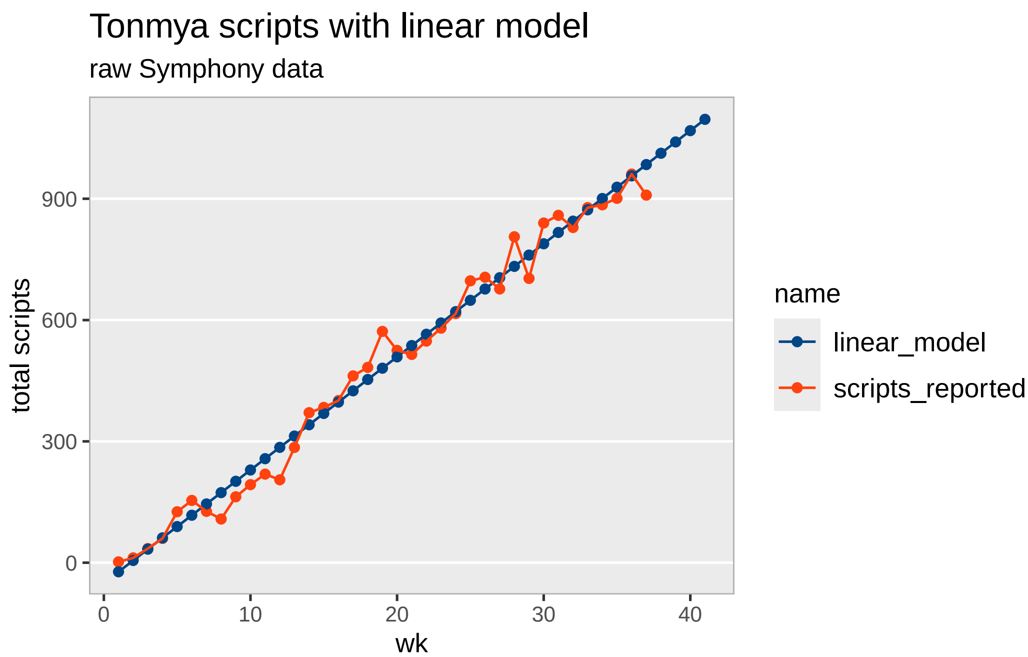
What are we doing here? - Just trying to get a grasp on the rate of increase and projecting out if that rate continues. It is not necessarily realistic to take it out as far as projections provide. Nothing here should be regarded as investment advice. I am not a financial analyst or advisor.

scripts Projection - at weekly rate - *linear model*

```
Call:
lm(formula = scripts ~ wk, data = df_scripts)

Residuals:
    Min      1Q  Median      3Q     Max
-80.194 -27.556   1.556  29.861  91.000

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -50.4730    13.4644  -3.749 0.000642 ***
wk           27.9723     0.6178  45.278  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40.12 on 35 degrees of freedom
Multiple R-squared:  0.9832,    Adjusted R-squared:  0.9827
F-statistic:  2050 on 1 and 35 DF,  p-value: < 2.2e-16
```

Scripts, Linear Model

wk	weekly_seq	total reported	linear predicted	annual
1	2025-11-14	2	-22.5	1,455
2	2025-11-21	12	5.5	2,909
3	2025-11-28	35	33.4	4,364
4	2025-12-05	60	61.4	5,818
5	2025-12-12	126	89.4	7,273
6	2025-12-19	154	117.4	8,727
7	2025-12-26	127	145.3	10,182
8	2026-01-02	108	173.3	11,636
9	2026-01-09	163	201.3	13,091
10	2026-01-16	193	229.2	14,546
11	2026-01-23	219	257.2	16,000
12	2026-01-30	205	285.2	17,455
13	2026-02-06	285	313.2	18,909
14	2026-02-13	371	341.1	20,364
15	2026-02-20	384	369.1	21,818
16	2026-02-27	401	397.1	23,273
17	2026-03-06	462	425.1	24,727
18	2026-03-13	483	453.0	26,182
19	2026-03-20	572	481.0	27,637
20	2026-03-27	525	509.0	29,091
21	2026-04-03	515	536.9	30,546
22	2026-04-10	548	564.9	32,000
23	2026-04-17	580	592.9	33,455
24	2026-04-24	616	620.9	34,909
25	2026-05-01	697	648.8	36,364
26	2026-05-08	706	676.8	37,818
27	2026-05-15	677	704.8	39,273
28	2026-05-22	806	732.8	40,728
29	2026-05-29	703	760.7	42,182
30	2026-06-05	840	788.7	43,637
31	2026-06-12	859	816.7	45,091
32	2026-06-19	829	844.6	46,546
33	2026-06-26	878	872.6	48,000
34	2026-07-03	885	900.6	49,455
35	2026-07-10	901	928.6	50,910
36	2026-07-17	961	956.5	52,364
37	2026-07-24	909	984.5	53,819
38	2026-07-31	NA	1,012.5	55,273
39	2026-08-07	NA	1,040.4	56,728
40	2026-08-14	NA	1,068.4	58,182
41	2026-08-21	NA	1,096.4	59,637

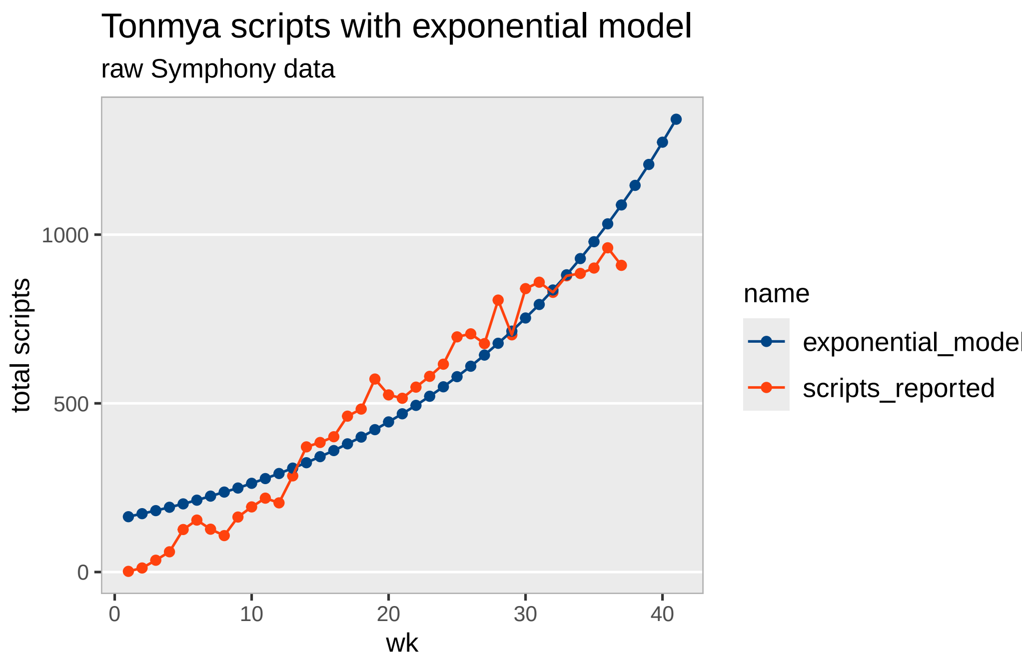


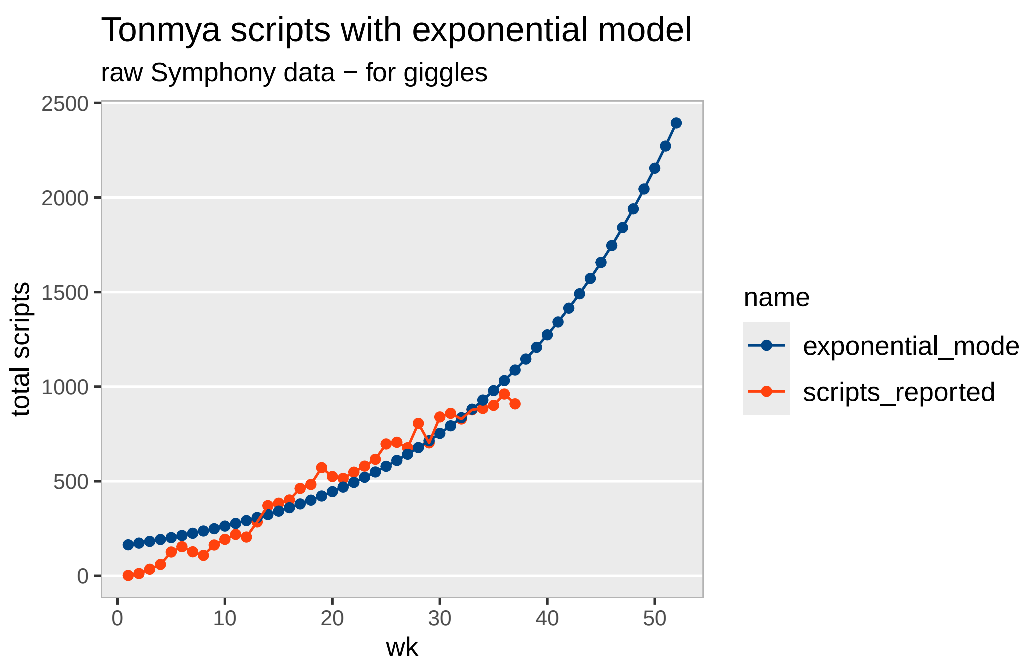
Scripts Projection - at weekly growth rate - *exponential model*

```
Formula: scripts ~ a * (1 + r)^wk

Parameters:
   Estimate Std. Error t value Pr(>|t|)
a 1.554e+02  1.633e+01   9.515 3.06e-11 ***
r 5.400e-02  3.717e-03  14.528  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 93.99 on 35 degrees of freedom

Number of iterations to convergence: 11
Achieved convergence tolerance: 3.879e-06
```

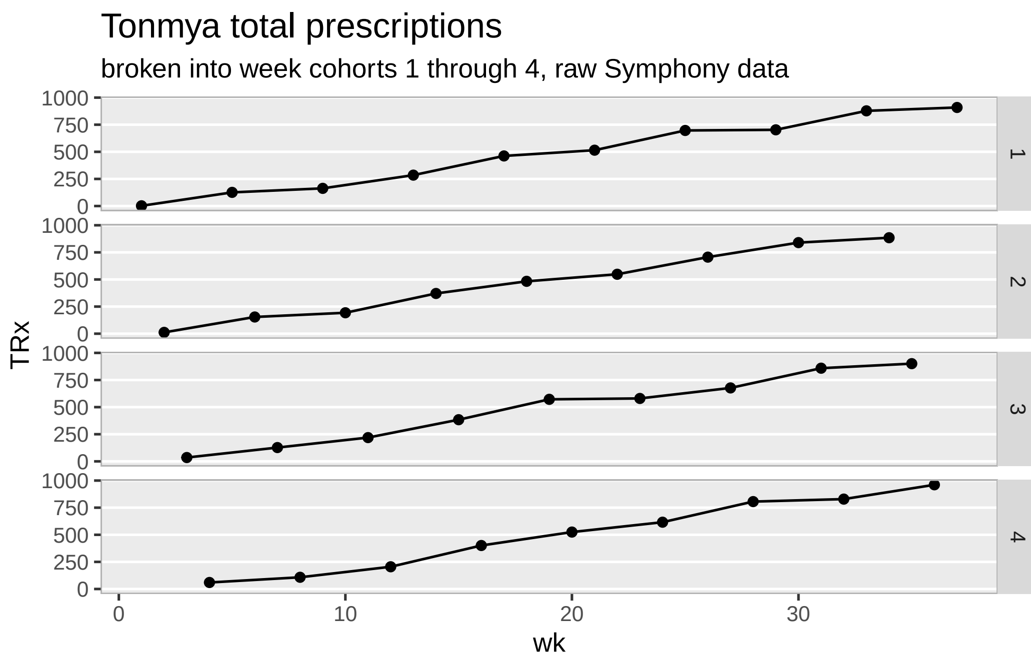
Scripts, Exponential Model

wk	weekly_seq	total reported	exp predicted	annual
1	2025-11-14	2	164	8,515
2	2025-11-21	12	173	8,975
3	2025-11-28	35	182	9,460
4	2025-12-05	60	192	9,971
5	2025-12-12	126	202	10,509
6	2025-12-19	154	213	11,077
7	2025-12-26	127	225	11,675
8	2026-01-02	108	237	12,305
9	2026-01-09	163	249	12,970
10	2026-01-16	193	263	13,670
11	2026-01-23	219	277	14,409
12	2026-01-30	205	292	15,187
13	2026-02-06	285	308	16,007
14	2026-02-13	371	324	16,871
15	2026-02-20	384	342	17,782
16	2026-02-27	401	360	18,743
17	2026-03-06	462	380	19,755
18	2026-03-13	483	400	20,822
19	2026-03-20	572	422	21,946
20	2026-03-27	525	445	23,132
21	2026-04-03	515	469	24,381
22	2026-04-10	548	494	25,697
23	2026-04-17	580	521	27,085
24	2026-04-24	616	549	28,548
25	2026-05-01	697	579	30,090
26	2026-05-08	706	610	31,715
27	2026-05-15	677	643	33,428
28	2026-05-22	806	678	35,233
29	2026-05-29	703	714	37,136
30	2026-06-05	840	753	39,141
31	2026-06-12	859	793	41,255
32	2026-06-19	829	836	43,483
33	2026-06-26	878	881	45,831
34	2026-07-03	885	929	48,306
35	2026-07-10	901	979	50,915
36	2026-07-17	961	1032	53,665
37	2026-07-24	909	1088	56,563
38	2026-07-31	NA	1146	59,618
39	2026-08-07	NA	1208	62,837
40	2026-08-14	NA	1274	66,231
41	2026-08-21	NA	1342	69,807





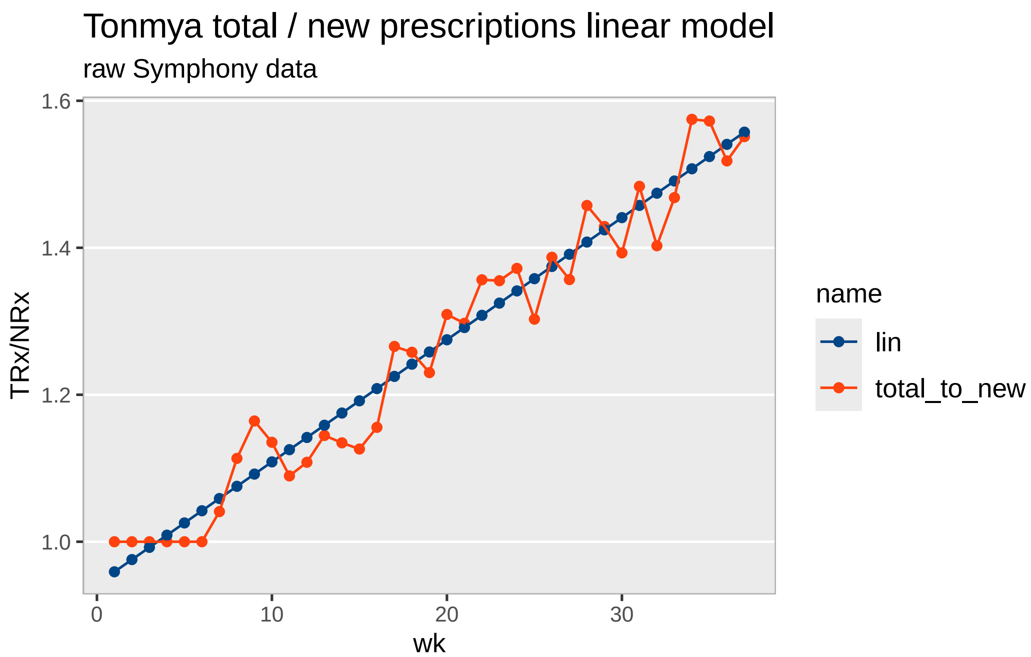
TRx Total Prescriptions by Weekly Cohorts



total / new scripts at weekly rate - *linear model*

```
Call:
lm(formula = total_to_new ~ wk, data = trx_nrx_df)

Residuals:
      Min        1Q    Median        3Q       Max
-0.071512 -0.033762  0.004495  0.030658  0.072268

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.9424607  0.0132899   70.92   <2e-16 ***
wk          0.0166175  0.0006098   27.25   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.0396 on 35 degrees of freedom
Multiple R-squared:  0.955, Adjusted R-squared:  0.9537
F-statistic: 742.6 on 1 and 35 DF,  p-value: < 2.2e-16
```

wk	scripts_reported
1	
1	2
5	126
9	163
13	285
17	462
21	515
25	697
29	703
33	878
37	909
2	
2	12
6	154
10	193
14	371
18	483
22	548
26	706
30	840
34	885
3	
3	35
7	127
11	219
15	384
19	572
23	580
27	677
31	859
35	901
4	
4	60
8	108
12	205
16	401
20	525
24	616
28	806
32	829
36	961

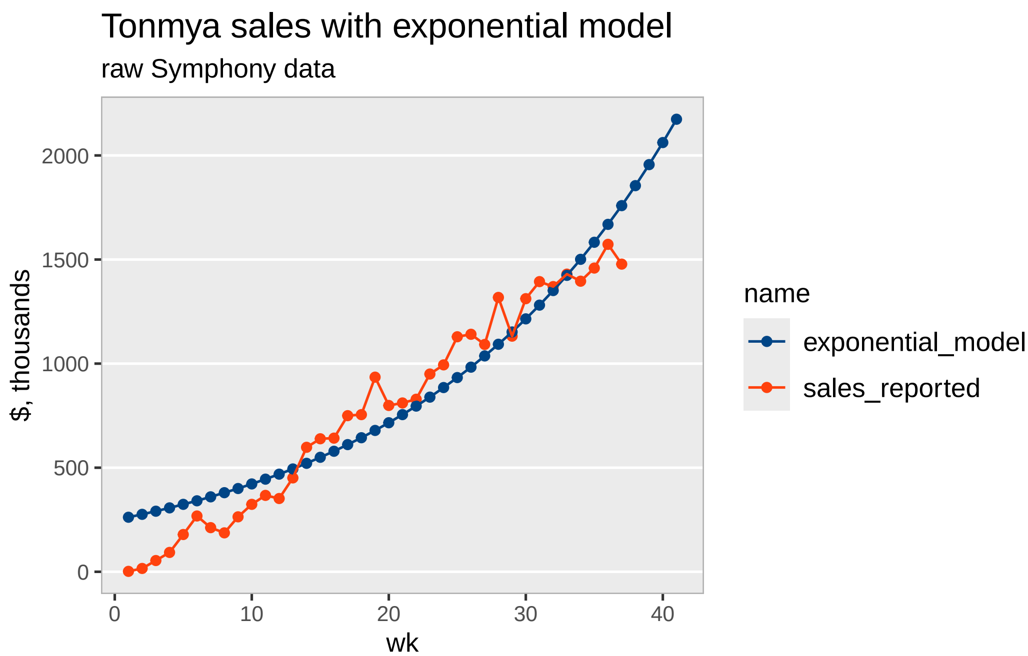


Sales Projection - at weekly growth rate - *exponential model*

```
Formula: sales ~ a * (1 + r)^wk

Parameters:
   Estimate Std. Error t value Pr(>|t|)
a 2.486e+02  2.578e+01   9.643 2.18e-11 ***
r 5.431e-02  3.665e-03  14.816  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 149.1 on 35 degrees of freedom

Number of iterations to convergence: 12
Achieved convergence tolerance: 8.21e-06
```

Exponential Model

wk	weekly_seq	sales, thousands	exp, thousands	annual, M
1	2025-11-14	$2	$262	$13.6
2	2025-11-21	$16	$276	$14.4
3	2025-11-28	$54	$291	$15.2
4	2025-12-05	$93	$307	$16.0
5	2025-12-12	$179	$324	$16.8
6	2025-12-19	$268	$341	$17.8
7	2025-12-26	$212	$360	$18.7
8	2026-01-02	$187	$380	$19.7
9	2026-01-09	$264	$400	$20.8
10	2026-01-16	$324	$422	$21.9
11	2026-01-23	$367	$445	$23.1
12	2026-01-30	$352	$469	$24.4
13	2026-02-06	$451	$494	$25.7
14	2026-02-13	$598	$521	$27.1
15	2026-02-20	$639	$550	$28.6
16	2026-02-27	$642	$579	$30.1
17	2026-03-06	$750	$611	$31.8
18	2026-03-13	$755	$644	$33.5
19	2026-03-20	$935	$679	$35.3
20	2026-03-27	$799	$716	$37.2
21	2026-04-03	$811	$755	$39.3
22	2026-04-10	$829	$796	$41.4
23	2026-04-17	$950	$839	$43.6
24	2026-04-24	$994	$885	$46.0
25	2026-05-01	$1,129	$933	$48.5
26	2026-05-08	$1,141	$983	$51.1
27	2026-05-15	$1,092	$1,037	$53.9
28	2026-05-22	$1,318	$1,093	$56.8
29	2026-05-29	$1,132	$1,152	$59.9
30	2026-06-05	$1,312	$1,215	$63.2
31	2026-06-12	$1,394	$1,281	$66.6
32	2026-06-19	$1,370	$1,351	$70.2
33	2026-06-26	$1,430	$1,424	$74.0
34	2026-07-03	$1,396	$1,501	$78.1
35	2026-07-10	$1,459	$1,583	$82.3
36	2026-07-17	$1,573	$1,669	$86.8
37	2026-07-24	$1,478	$1,759	$91.5
38	2026-07-31	NA	$1,855	$96.5
39	2026-08-07	NA	$1,956	$101.7
40	2026-08-14	NA	$2,062	$107.2
41	2026-08-21	NA	$2,174	$113.0



Tonmya sales with exponential model
raw Symphony data
2000
1500
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$, thousands
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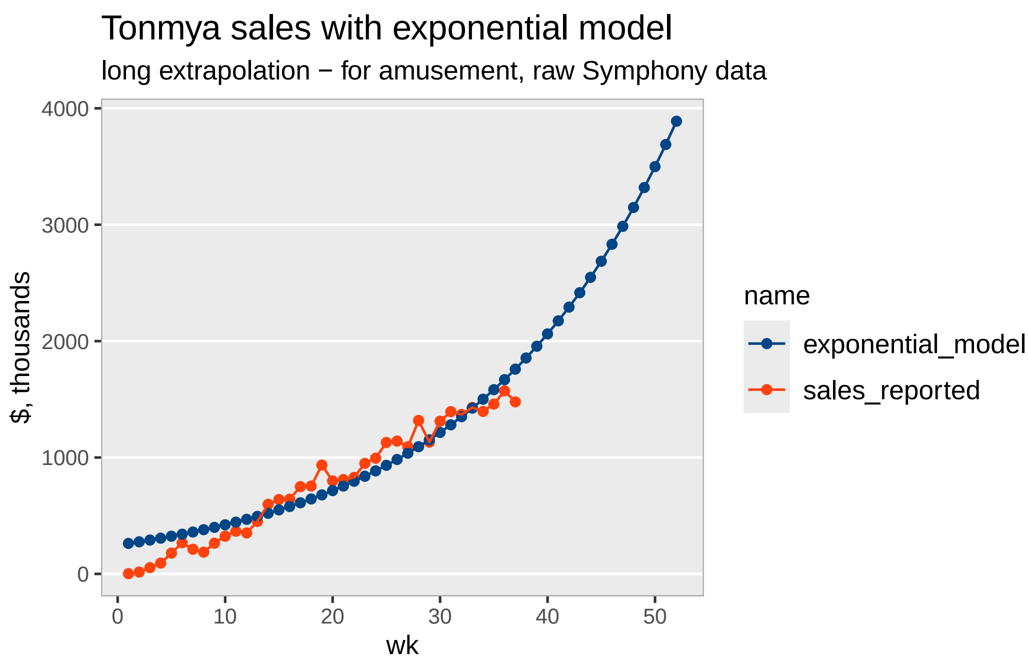
30
40
wk
name
exponential_model
sales_reported



Tonmya sales with exponential model
long extrapolation − for amusement, raw Symphony data
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$, thousands
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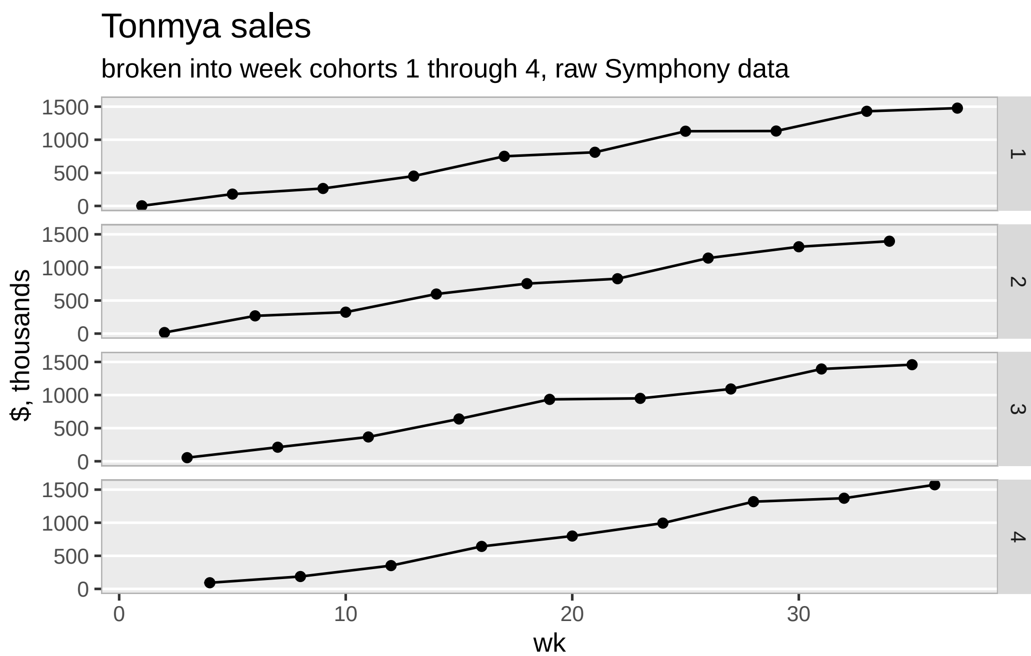
20
30
40
50
wk
name
exponential_model
sales_reported

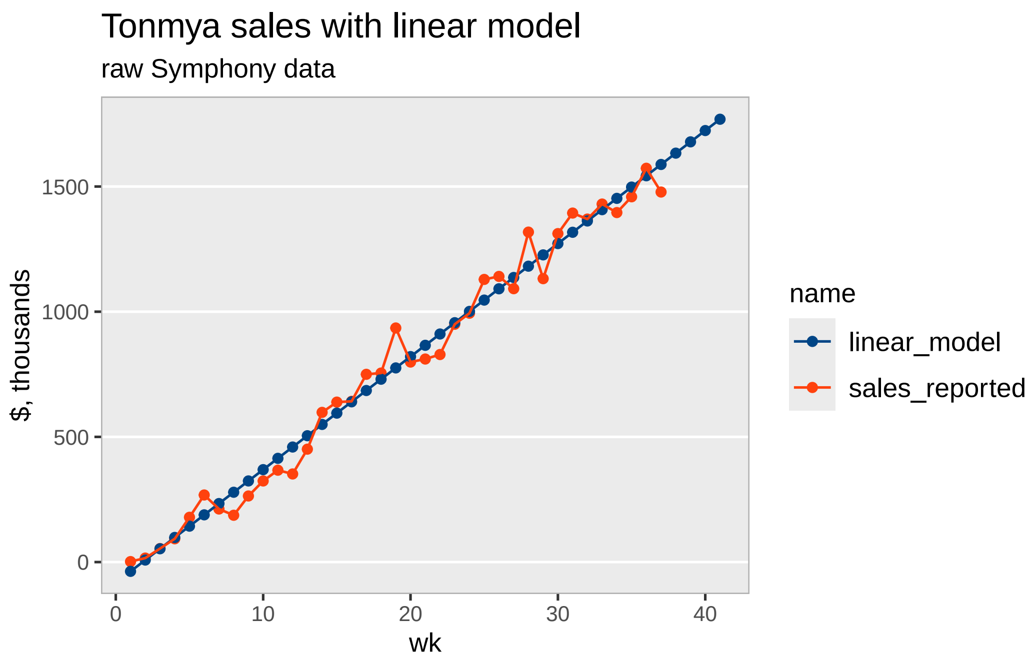


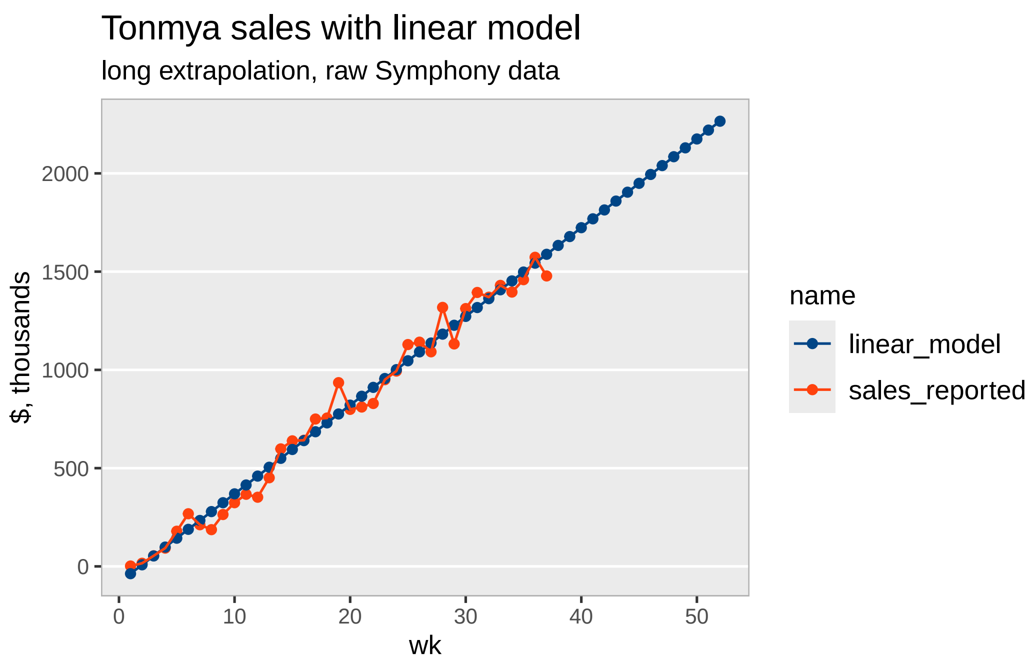
Sales Projection - at weekly rate - *linear model*

```
Call:
lm(formula = sales ~ wk, data = df_sales)

Residuals:
     Min       1Q   Median       3Q      Max
-110.182  -47.367    0.807   39.845  159.459

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -82.248     21.957  -3.746 0.000647 ***
wk            45.147      1.007  44.812  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 65.43 on 35 degrees of freedom
Multiple R-squared:  0.9829,    Adjusted R-squared:  0.9824
F-statistic:  2008 on 1 and 35 DF,  p-value: < 2.2e-16
```

Linear Model

wk	weekly_seq	sales, thousands	linear, thousands	annual, M
1	2025-11-14	$2	-$37	$2.3
2	2025-11-21	$16	$8	$4.7
3	2025-11-28	$54	$53	$7.0
4	2025-12-05	$93	$98	$9.4
5	2025-12-12	$179	$143	$11.7
6	2025-12-19	$268	$189	$14.1
7	2025-12-26	$212	$234	$16.4
8	2026-01-02	$187	$279	$18.8
9	2026-01-09	$264	$324	$21.1
10	2026-01-16	$324	$369	$23.5
11	2026-01-23	$367	$414	$25.8
12	2026-01-30	$352	$460	$28.2
13	2026-02-06	$451	$505	$30.5
14	2026-02-13	$598	$550	$32.9
15	2026-02-20	$639	$595	$35.2
16	2026-02-27	$642	$640	$37.6
17	2026-03-06	$750	$685	$39.9
18	2026-03-13	$755	$730	$42.3
19	2026-03-20	$935	$776	$44.6
20	2026-03-27	$799	$821	$47.0
21	2026-04-03	$811	$866	$49.3
22	2026-04-10	$829	$911	$51.6
23	2026-04-17	$950	$956	$54.0
24	2026-04-24	$994	$1,001	$56.3
25	2026-05-01	$1,129	$1,046	$58.7
26	2026-05-08	$1,141	$1,092	$61.0
27	2026-05-15	$1,092	$1,137	$63.4
28	2026-05-22	$1,318	$1,182	$65.7
29	2026-05-29	$1,132	$1,227	$68.1
30	2026-06-05	$1,312	$1,272	$70.4
31	2026-06-12	$1,394	$1,317	$72.8
32	2026-06-19	$1,370	$1,362	$75.1
33	2026-06-26	$1,430	$1,408	$77.5
34	2026-07-03	$1,396	$1,453	$79.8
35	2026-07-10	$1,459	$1,498	$82.2
36	2026-07-17	$1,573	$1,543	$84.5
37	2026-07-24	$1,478	$1,588	$86.9
38	2026-07-31	NA	$1,633	$89.2
39	2026-08-07	NA	$1,678	$91.6
40	2026-08-14	NA	$1,724	$93.9
41	2026-08-21	NA	$1,769	$96.3





Refills - at weekly rate - *linear model*

```
Call:
lm(formula = refills ~ wk, data = trx_nrx_df)
```

```
Residuals:
    Min      1Q  Median      3Q     Max 
-41.267 -19.153  -7.606  17.431  58.110 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -68.2793     9.2255  -7.401 1.17e-08 ***
wk           10.1697     0.4233  24.025  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 27.49 on 35 degrees of freedom
Multiple R-squared:  0.9428,	Adjusted R-squared:  0.9412 
F-statistic: 577.2 on 1 and 35 DF,  p-value: < 2.2e-16
```